Homework 4

Solutions

Exercise 4.1 (Hogg, Tanis and Zimmerman, 5.1-6)**.** Let X have **logistic distribution** with probability density function

$$f(x) = \frac{e^{-x}}{(1+e^{-x})^2}, \qquad -\infty < x < \infty.$$

Show that

$$Y = \frac{1}{1+e^{-X}}$$

has a $\text{Unif}(0,1)$ distribution.

Proof. We first observe that for any $0 < t < 1$,

$$F_Y(t) = \mathbb{P}(Y \le t) = \mathbb{P}\left(\frac{1}{1+e^{-X}} \le t\right).$$

Rearranging to solve for X,

$$\begin{aligned}\mathbb{P}\left(\frac{1}{1+e^{-X}} \le t\right) = \mathbb{P}\left(1+e^{-X} \ge \frac{1}{t}\right) = \mathbb{P}\left(e^{-X} \ge \frac{1}{t} - 1\right) = \mathbb{P}\left(e^{-X} \ge \frac{1-t}{t}\right) \\ = \mathbb{P}\left(e^{X} \le \frac{t}{1-t}\right) = \mathbb{P}\left(X \le \ln\left(\frac{t}{1-t}\right)\right).\end{aligned}$$

Now, we could differentiate immediately to find the pdf of X, but the algebra is a bit annoying. Instead, notice that

$$\mathbb{P}\left(X \le \ln\left(\frac{t}{1-t}\right)\right) = \int_{-\infty}^{\ln\left(\frac{t}{1-t}\right)} \frac{e^{-x}}{(1+e^{-x})^2}\, dx.$$

Using a u-substitution with $u = 1+e^{-x}$, we find

$$\int_{-\infty}^{\ln\left(\frac{t}{1-t}\right)} \frac{e^{-x}}{(1+e^{-x})^2}\, dx = \int_{\infty}^{1+\frac{1-t}{t}} \frac{-du}{u^2} = \int_{1/t}^{\infty} \frac{du}{u^2} = \frac{-1}{u}\Big|_{1/t}^{\infty} = t.$$

For $t \le 0$, $F_Y(t) = 0$ since Y cannot be negative, and for $t \ge 1$, $F_Y(t) = 1$ since Y is always between 0 and 1. So,

$$\boxed{F_Y(t) = \begin{cases} 0 & \text{if } t \le 0 \\ t & \text{if } 0 < t < 1 \\ 1 & \text{if } 1 \le t. \end{cases}}$$

Differentiating, we see that $f_Y(t) = \mathbf{1}_{(0,1)}$, which means that $Y \sim \text{Unif}(0,1)$. □

Exercise 4.2. Let $X_1, \ldots, X_n \overset{i.i.d}{\sim} \text{Ber}(p)$. Compute the moment generating function of $X_1 + \cdots + X_n$, and conclude that it has a $\text{Bin}(n,p)$ distribution.

Sol.

Recall that the mgf of a $\text{Ber}(p)$ random variable is given by

$$1 - p + pe^t.$$

By independence of each X_i, we have

$$M_{X_1+\cdots+X_n}(t) = \mathbb{E}\left[e^{t(X_1+\cdots+X_n)}\right] = \mathbb{E}\left[e^{tX_1}\cdots e^{tX_n}\right] = \prod_{i=1}^{n} \mathbb{E}\left[e^{tX_i}\right] = (1-p+pe^t)^n.$$

This is precisely the mgf of a $\text{Bin}(n,p)$ random variable, so we conclude that $X_1 + X_2 + \cdots + X_n \sim \text{Bin}(n,p)$.

Exercise 4.3 (Panaretos, Exercise 22 (edited))**.** Let $X_1, \dots, X_n \overset{i.i.d}{\sim} \text{Poisson}(\lambda)$.

1. Show that $T(X_1, \dots, X_n) = \frac{1}{n}\sum_{i=1}^n X_i$ is a sufficient statistic for λ.
2. Find the sampling distribution of T.

Sol. We use the Fisher-Neyman factorisation. Observe from independence that the joint probability mass function can be expressed, for any $\vec{k} = (k_1, \dots, k_n) \in \{0, 1, 2, 3 \dots\}^n$,

$$f_{X_1,\dots,X_n}(k_1, \dots, k_n) = \prod_{i=1}^n f_{X_i}(k_i) = \prod_{I=1}^n \left(e^{-\lambda}\frac{\lambda^{k_i}}{k_i!}\right) = e^{-n\lambda}\frac{\lambda^{\sum_{i=1}^n k_i}}{\prod_{i=1}^n k_i!} = e^{-n\lambda}\frac{\lambda^{nT(\vec{k})}}{\prod_{i=1}^n k_i!}.$$

Defining $g(T, \lambda) = e^{-n\lambda}\lambda^{nT}$ and $h(k_1, \dots, k_n) = \frac{1}{\prod_{i=1}^n k_i!}$, we see that

$$f_{X_1,\dots,X_n}(k_1, \dots, k_n) = g(T(k_1, \dots, k_n), \lambda)h(k_1, \dots, k_n),$$

which is the desired factorisation. Hence, T is sufficient for λ.

To find the sampling distribution, we must compute

$$\mathbb{P}(T \le t) = \mathbb{P}\left(\frac{X_1 + X_2 + \cdots + X_n}{n} \le t\right) = \mathbb{P}(X_1 + X_2 + \cdots + X_n \le nt).$$

Instead of arguing directly from this point, recall from HW 1 that $\sum_{i=1}^n X_i \sim \text{Poisson}(n\lambda)$. Hence,

$$\mathbb{P}(T \le t) = \sum_{k=0}^{nt} \mathbb{P}(X_1 + X_2 + \cdots + X_n = k) = \sum_{k=0}^{nt} \mathbb{P}(\text{Poisson}(n\lambda) = k) = \boxed{\sum_{k=0}^{nt} e^{-\lambda}\frac{\lambda^k}{k!}}.$$